

REPORT ON FIRST YEAR INTENSHIP-2025

Carried out on

AUTOMATIC STREET LIGHT CIRCUIT

Submitted to

NMAM INSTITUTE OF TECHNOLOGY, NITTE

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In partial fulfillment of the requirements for the award of the

Degree of Bachelor of Technology

In

Electronics and Communication Engineering

By

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OF TECHNOLOGY**

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ABSTRACT

This project presents the design and implementation of an automatic street light system aimed at enhancing energy efficiency and safety in urban environments. The system ensures that street lights are only activated during low-light conditions, conserving energy during daylight hours. Through this mini project, the integration of smart technologies offers a practical solution for sustainable urban lighting management, contributing to a more efficient and safer public infrastructure.

PROJECT DETAILS

Introduction

The demand for energy-efficient solutions in urban infrastructure has led to the development of innovative technologies aimed at optimizing resource consumption. Among these, automatic street light circuits have gained prominence for their ability to enhance energy efficiency in public lighting systems. This mini project focuses on the design and implementation of an automatic street light circuit, leveraging Light Dependent Resistors (LDRs) for ambient light detection. Unlike traditional systems that rely on manual switches or timers, this circuit offers a dynamic approach by adjusting street light illumination based on real-time variations in natural light levels. By utilizing LDRs as sensors, the circuit ensures that street lights are activated only when necessary, minimizing energy wastage during periods of sufficient natural

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illumination. Through this project, we aim to contribute to the development of sustainable urban lighting solutions that not only conserve energy but also enhance safety and visibility in public spaces.

Backend Theory

The automatic street light circuit's ability to operate efficiently and independently depends on the interaction of its individual parts. The principal sensor, the Light Dependent Resistor (LDR), senses changes in the surrounding light levels. The resistance of the LDR drops in dimmed environments, changing the voltage across its terminals in response. The LM324 operational amplifier integrated circuit (IC) amplifies and processes this voltage modification before comparing it to a reference voltage established by the potentiometer and resistor combination. The operational amplifier completes the circuit by. This allows the 9V battery to power the LED strip. As a result, the LED strip lights up to resemble street lights.

Methodology Used

In addition to being battery-operated, this circuit is quite simple to work around. Because there are so few components in the circuit, the circuit uses relatively little electricity

. The IC LM324-basically an operational amplifier housed in a voltage comparator is the foundation of the entire circuit. The primary element for light sensing is an LDR (light-dependent resistor), who resistance

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depends on the amount of light falling on it. A few additional components are also utilized in addition to these. Understanding how a circuit operates is rather simple.

We utilized IC LM324 in this circuit, which functions essentially as an operational amplifier. These integrated circuits' pins 2 and 3 are used to compare voltages and provide an output that is either high or low based on the voltages at the input pins, in this circuit, LDR and 10KQ Resistor form one potential divider pair.

Which is used to provide a variable voltage at the non-inverting input (that is Pin 3). A 10K potentiometer is used to supply half of the supply voltage to the inverting pin, which is the center of the second potential divider.

Since the LDR has a low resistance during the day, as we know, the voltage pin 3, which is the non-inverting input, is higher than the voltage at pin 2. Which is the inverting input. The output at pin 1 is hence high. Consequently, neither the LED nor the bulb will glow.

However, we are aware that the resistance of the LDR is strong in low light or at night. As a result, the IC LM324's non-inverting input pin 3 has a lower voltage than its inverting input pin 2. Consequently, the output pin 1 enters a low condition, further activating the relay and causing the LED or bulb connected to it to shine. Ground is pin11, Vcc is pin4.

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Components required:

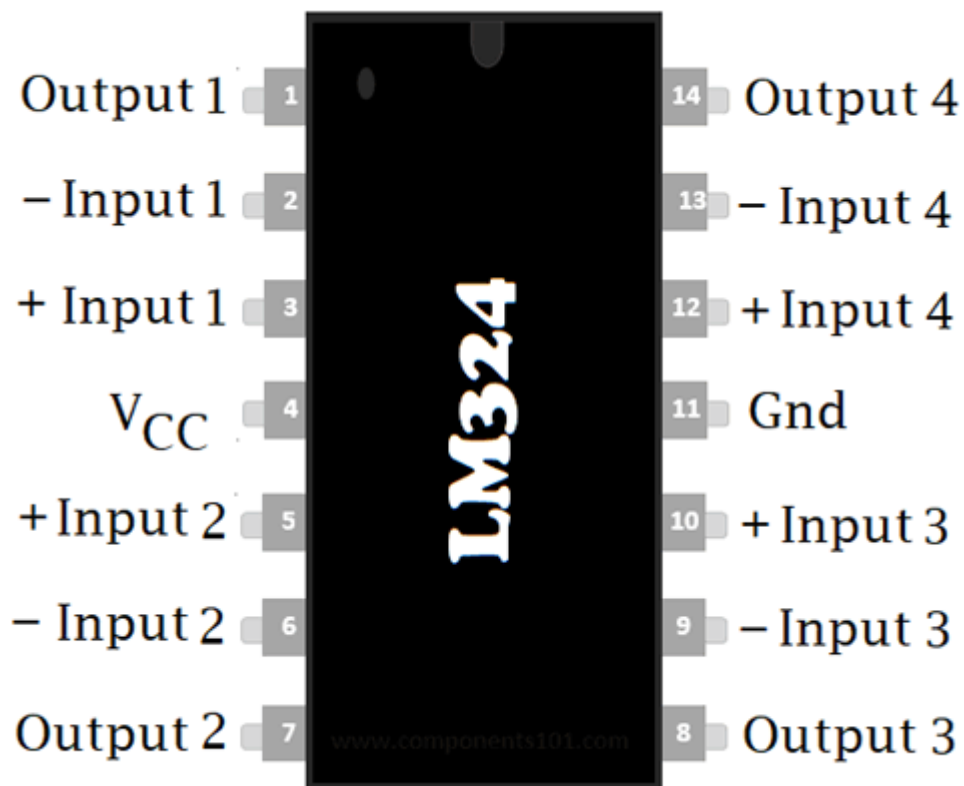
S NO.	NAME OF THE COMPONENT	SPECIFICATIONS	QUANTITY
1	Bread board	-	1
2	IC	LM324	1
3	Resistor	10k ohm	1
4	Potentiometer	10k ohm	1
5	HW-131	3.3v-5v	1
6	Battery	9v	1
7	small LED bulb	-	1
8	LDR	-	1
9	connecting wires	-	

Expected results:

As we know the property of LDR that amid the day time, its resistance is low, the voltage at the non-inverting input (i.e. pin 3) is higher than the voltage at the modifying input (pin 2) consequently, the yield at the pin 1 is high. As a result, the Driven (or the bulb) will not shine. But in obscurity or at night time, we know that resistance of LDR is high. Thus, the voltage at non-inverting input stick 3 of the IC LM324 diminishes than the rearranging input pin 2. As a result, the output pin i moves to low state, the LED or bulb associated in a will glow.

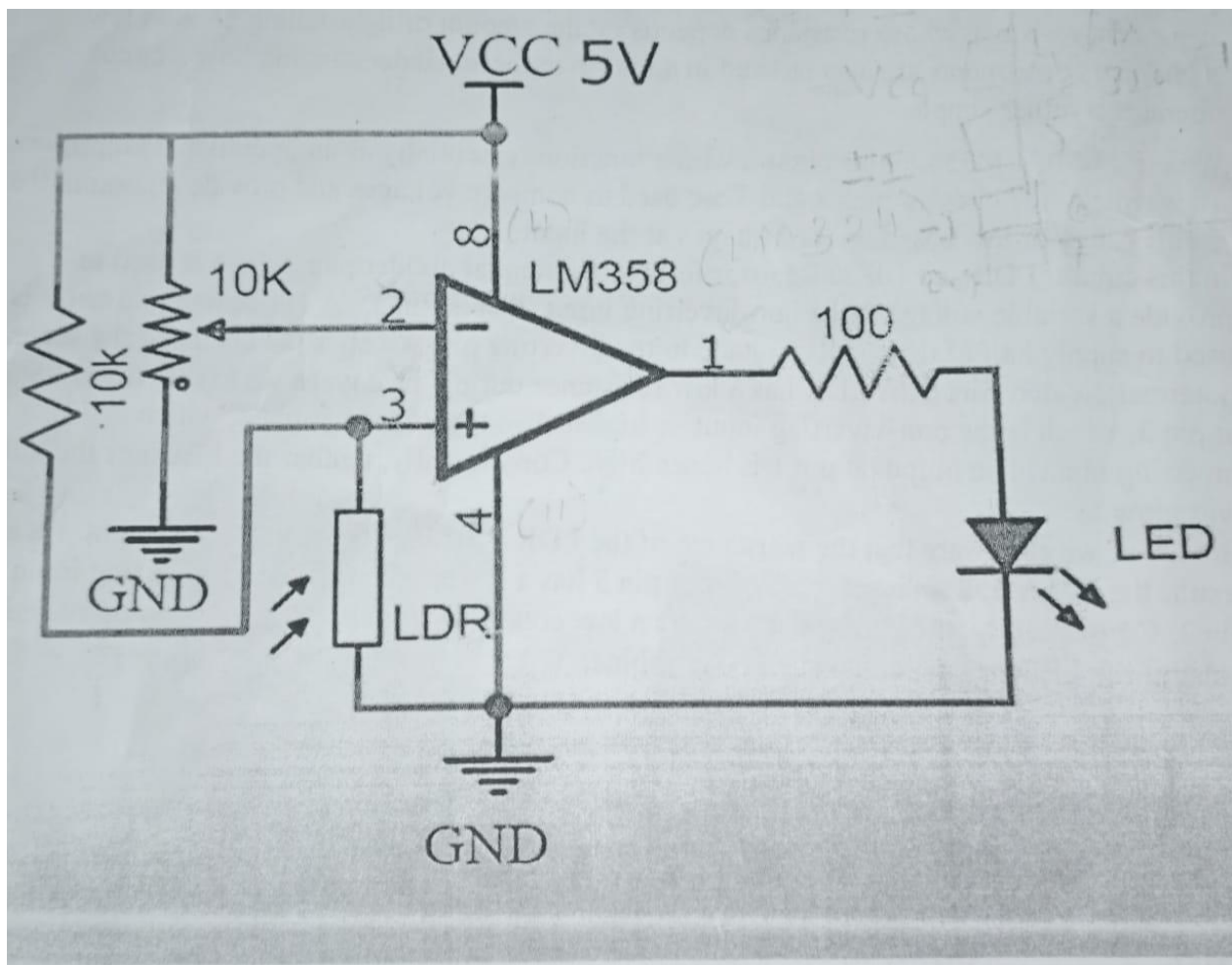
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Pin diagram:



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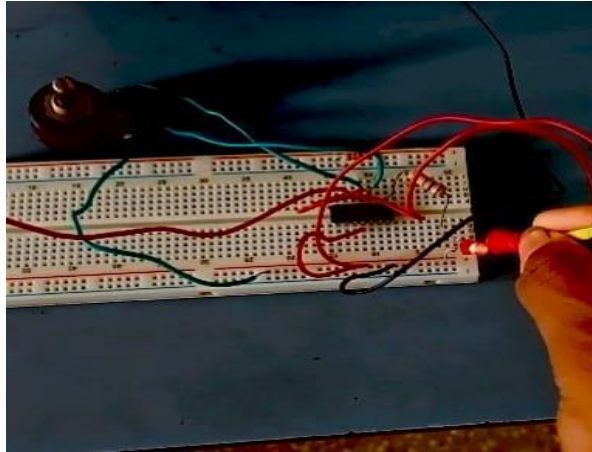
Circuit diagram:



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Progress mode:

(During Daytime Light will not glow)



(During Night time light will glow)



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Applications

Beyond its intended use, the small project on autonomous street light circuits has a plethora of additional uses. First of all, because of its energy-efficient design, it can be used in smart city projects where minimizing resource usage is crucial. Furthermore, outdoor lighting systems in parks, residential areas, and public places benefit greatly from the circuit's capacity to adjust street light intensity based on ambient light levels, which improves safety and security while saving energy. The idea can also be expanded to other fields, like agricultural automation, where it can regulate lighting in greenhouses to mimic natural day night cycles and promote the best possible growth for plants. Additionally, because of its affordability and ease of use, it can be used in education to help students understand the foundational ideas of automation and electronics.

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